

Question ID: 8643d906

| Assessment | Test | Domain  | Skill            | Difficulty                                        |
|------------|------|---------|------------------|---------------------------------------------------|
| SAT        | Math | Algebra | Linear functions | Easy <div><div></div><div></div><div></div></div> |

Question  
 $P(t) = 250 + 10t$

The population of snow leopards in a certain area can be modeled by the function P defined above, where  $P(t)$  is the population t years after 1990. Of the following, which is the best interpretation of the equation  $P(30) = 550$  ?

- Answer
- A. The snow leopard population in this area is predicted to be 30 in the year 2020.
  - B. The snow leopard population in this area is predicted to be 30 in the year 2030.
  - C. The snow leopard population in this area is predicted to be 550 in the year 2020.
  - D. The snow leopard population in this area is predicted to be 550 in the year 2030.

Correct Answer: C

Rationale

Choice C is correct. It’s given that  $P(t)$  represents the population of snow leopards t years after 1990.  $P(30) = 550$  corresponds to  $t = 30$  and  $P(t) = 550$ . It follows that  $t = 30$  corresponds to 30 years after 1990, or 2020. Thus, the best interpretation of  $P(30) = 550$  is that the snow leopard population in this area is predicted to be 550 in the year 2020.

Choices A and B are incorrect and may result from reversing the interpretations of t and  $P(t)$ . Choice D is incorrect and may result from determining that 30 years after 1990 is 2030, not 2020.